THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 6: **Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 21	each term is the sum of the two before it
2	**(b)** odd	odd + even
3	**(a)** odd	odd + even
4	**(a)** odd	a product is odd only if both numbers are odd
5	**(c)** 1234	reads differently from the right
6	**(b)** 0 or 5	rule
7	**(a)** n^2	a square-number pattern
8	**(d)** 5	3 × 5 = 15
9	**(d)** rhythm patterns (long and short beats)	poetic metres
10	**(c)** 1,254	digits read backwards

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The sum is even (1); 7 + 9 = 16 (1).	1+1
12	Odd (1); no factor 2 is present (1).	1+1
13	1, 2, 3, 5, 8, 13, 21, 34 (2).	2
14	A number that reads the same both ways (1); 121 and 4554 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	n + (n + 1) + (n + 2) = 3n + 3 (1.5); = 3(n + 1) (1.5).	1½+1½
16	Possible with 1 and 2 beats: 13 ways (3).	3
17	5432 − 2345 = 3087 (3).	3
	OR: e.g. 25 + 25 = 50 (A=2, B=5, C=0) (3).	3
18	2 7 6 / 9 5 1 / 4 3 8 (3).	3
	OR: 7 + 2 + 8 + 4 = 21, divisible by 3 (1.5); 84 divisible by 4 (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 1, 2, 3, 5, 8, 13, 21, 34, 55, 89 (2). (ii) Add previous two (1). (iii) 11th term (2).	2+1+2
	OR: (i) 5 (1). (ii) 1111, 112, 121, 211, 22 (2). (iii) 8 (1). (iv) Virahāṅka (1).	1+2+1+1
20	(i) (2a + 1) + (2b + 1) = 2(a + b + 1) (1.5). (ii) (2a + 1)(2b + 1) = 2(…) + 1 (1.5). (iii) 63 + 11 = even (2).	1½+1½+2
	OR: (i) 4321 − 1234 = 3087 (1). (ii) 8730 − 378 = 8352; 8532 − 2358 = 6174 (3). (iii) 6174 (1).	1+3+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 6: **Number Play** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 34	each term is the sum of the two before it
2	**(a)** odd	even + odd
3	**(b)** odd	a product is odd only if both numbers are odd
4	**(a)** even	odd + odd
5	**(b)** 6174	repeated rearrangement process
6	**(d)** 4	100 is divisible by 4
7	**(c)** 20	2n
8	**(d)** 5	5 + 5
9	**(a)** 8	Virahāṅka number
10	**(b)** 4	6 × 4 = 24

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	3 + 6 + 5 + 4 = 18 (1); yes (1).	1+1
12	36 (2).	2
13	Even, as it ends in 6 (2).	2
14	15; 5 (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$10^2 = 100$ (1.5); pattern n^2 (1.5).	1½+1½
16	2a + 2b = 2(a + b) (3).	3
17	9, 18, 27 … digits add to 9 (3).	3
	OR: 77 (1.5); palindrome (1.5).	1½+1½
18	5 + x + 3 divisible by 9: x = 1 (3).	3
	OR: Even; it has the even factors 14 (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 15 (1). (ii) Even numbers 2, 4, 6, 8 (1). (iii) 2 7 6 / 9 5 1 / 4 3 8 (2). (iv) 2 + 5 + 8 = 15 (1).	1+1+2+1
	OR: (i) Last digit even; digit sum by 3; last digit 0/5; digit sum by 9; last digit 0 (2). (ii) Divisible by 2, 3, 5, 9, 10 (1.5). (iii) By 3, 5, 9 (1.5).	2+1½+1½
20	(i) A = 8, B = 2 (4). (ii) 24 + 4 = 28 (1).	4+1
	OR: (i) 4^2 (1). (ii) 144 (2). (iii) Each L-shaped layer adds the next odd number (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.